

BATUMI, Georgia

WMO: 374840

Lat: 41.610N

Lon: 41.600E

Elev: 22

StdP: 14.68

Time Zone: 4.00 (E04)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.8	33.6	19.8	14.9	44.6	22.7	17.1	44.7	32.2	49.0	29.9	49.7	11.1	140	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.3	85.9	76.7	84.2	76.1	82.5	74.9	78.8	84.3	77.5	82.9	76.2	81.4	8.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.9	140.4	83.7	75.4	133.3	81.8	74.0	127.1	80.3	42.4	84.3	40.7	83.3	39.6	81.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.6	25.1	22.5	DB	27.7	93.3	3.4	3.3	25.2	95.7	23.3	97.6	21.4	99.5	18.9	101.9
			WB	26.1	79.9	3.8	1.2	23.4	80.8	21.2	81.5	19.1	82.2	16.3	83.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	46.8	47.3	50.8	55.4	62.9	70.5	74.3	76.3	70.6	63.5	55.8	49.8
	DBStd	11.92	7.10	7.21	7.80	6.82	5.68	3.99	3.40	3.40	4.41	5.32	6.59	7.51
	HDD50	498	147	129	85	16	0	0	0	0	0	1	23	97
	HDD65	2772	564	497	446	299	106	4	0	0	10	92	282	472
	CDD50	4303	49	53	109	179	401	616	755	817	618	418	197	91
	CDD65	1102	0	0	5	13	42	170	290	352	178	44	6	2
	CDH74	6172	0	1	54	97	165	704	1715	2537	773	109	13	4
	CDH80	1103	0	0	11	22	53	89	274	565	72	17	0	0
(14) Wind	WSAvg	10.1	13.4	11.4	10.7	8.8	7.7	7.9	7.8	7.9	9.2	10.0	12.4	13.5
(15) Precipitation	PrecAvg	101.4	9.9	7.8	6.1	4.3	3.6	6.7	6.7	9.0	11.6	11.5	11.2	12.0
	PrecMax	137.5	15.9	16.5	10.5	12.2	7.4	13.1	14.7	21.4	19.9	23.5	25.4	22.9
	PrecMin	79.4	3.7	1.2	2.1	0.8	1.1	2.9	2.8	1.4	3.8	0.4	0.4	4.3
	PrecStd	12.6	3.7	3.8	2.0	2.3	1.7	2.4	2.9	5.0	4.0	5.6	6.3	4.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.0	70.6	80.7	83.1	86.3	86.3	86.4	88.1	84.4	82.6	75.5	73.0
		MCWB	51.0	52.9	58.3	61.1	68.9	73.2	78.2	79.7	75.8	67.0	58.2	53.9
	2%	DB	63.0	64.8	73.2	76.2	77.1	82.2	84.4	86.1	82.1	76.8	71.4	67.8
		MCWB	48.2	51.2	55.6	59.6	66.9	74.9	76.9	78.4	73.8	66.4	57.0	51.3
	5%	DB	59.3	61.0	68.1	70.9	73.6	80.3	82.7	84.5	80.3	73.7	67.7	63.7
		MCWB	47.0	48.7	53.3	57.6	65.4	73.4	75.8	77.3	72.7	65.7	56.3	50.2
	10%	DB	55.8	57.4	62.8	65.9	71.5	78.1	81.0	83.1	78.7	71.5	64.5	60.4
		MCWB	46.1	47.4	51.7	55.9	64.6	71.9	74.4	76.2	71.7	64.6	55.1	49.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	55.2	59.7	63.0	71.6	77.4	79.8	80.2	76.9	70.5	62.7	57.4
		MCDB	60.6	66.3	76.5	76.6	79.4	82.3	84.9	86.5	82.9	76.3	69.2	62.9
	2%	WB	50.8	52.2	56.4	60.4	68.5	75.5	78.3	79.2	74.9	68.5	60.1	55.0
		MCDB	57.5	61.5	70.8	72.8	74.9	81.1	83.4	85.1	80.4	74.6	66.3	61.6
	5%	WB	49.4	50.4	54.2	58.6	66.9	73.9	76.7	78.2	73.3	66.8	58.5	52.9
		MCDB	55.9	57.5	65.5	67.9	72.6	79.3	81.8	83.9	78.7	72.5	64.3	59.7
	10%	WB	48.1	48.9	52.2	56.8	65.2	72.2	75.4	77.0	71.9	65.2	56.7	51.0
		MCDB	55.0	54.8	61.0	64.4	70.5	77.0	80.4	82.4	77.4	70.4	62.8	58.0
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.0	13.3	13.0	11.2	11.4	10.7	11.3	12.0	12.0	12.5	11.5
		MCDBR	15.2	17.7	22.9	22.4	16.4	13.8	12.1	12.2	13.5	14.9	16.5	15.3
		MCWBR	8.2	9.2	10.5	10.7	10.4	9.5	8.2	8.3	9.7	10.7	10.5	8.6
		MCDBR	13.0	15.5	20.1	19.0	14.3	12.7	11.5	11.6	12.7	13.6	13.9	12.8
	5% WB	MCWBR	8.1	8.5	9.7	10.1	9.8	9.0	8.1	8.1	9.8	10.3	10.4	8.5
(40) Clear-Sky Solar Irradiance	taub		0.353	0.387	0.443	0.496	0.462	0.443	0.470	0.481	0.422	0.419	0.368	0.345
	taud		2.284	2.175	2.061	1.991	2.181	2.306	2.264	2.223	2.369	2.294	2.376	2.344
	Ebn at Noon		245	253	252	247	260	265	257	250	258	241	240	238
	Edh at Noon		29	37	47	54	46	41	42	43	35	33	27	25
(44) All-Sky Solar Radiation	RadAvg		507	751	1040	1437	1720	1865	1763	1605	1349	917	633	452
	RadStd		66	89	95	169	145	155	176	136	119	101	92	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-198	+173	+68

Nomenclature: See separate page